

Replication Report: *Genome sequencing and comparative genomic analysis of bovine mastitis-associated Staphylococcus aureus strains from India* (Sivakumar et al., BMC Genomics 24:44, 2023)

Automated replication via BV-BRC API
Backfill: 2026-07-05

Original replication: 2026-05-05 Backfill to 8-artifact standard: 2026-07-05

Abstract

Sivakumar et al. (2023) present the first whole-genome-sequencing (WGS) and comparative genomic characterization of 41 bovine mastitis-associated *Staphylococcus aureus* strains isolated in India. Their central results are: 15 sequence types (STs) resolving to 5 clonal complexes (CCs) dominated by CC8 (21) and CC97 (10); all 41 are methicillin-susceptible (MSSA); ~2.7 Mbp mean genome, 32.7% GC; a Roary+BPGA pan-genome of 4,360 genes with a 1,878-gene core, described as “almost closed” ($b = 0.0817389$); an SNP tree of six major clades; 16 spa types; 17 AMR genes (mostly efflux, plus *blaZ* in 14, *tet(K)* in 1, *TEM-116* and *TEM-181* in one strain each); 108 virulence genes with clade-specific distribution of enterotoxins, PVL confined to strain A3.1 (ST243), and *tsst* in 14 strains. We replicated the study end-to-end from public data using the BV-BRC API on 41/41 strains (100% scope, 31/33 testable claims tested). Every genome-count-based claim reproduces; three quantitative totals (pan-genome size, VF count, AMR gene count) differ by expected amounts because we used PLFam clustering (not Roary), VFDB via BV-BRC (not the paper’s VFanalyzer version), and CARD via BV-BRC (which includes more intrinsic resistance genes). Two claims are not testable in BV-BRC (*spa* typing and Roary-based power-law pan-genome closure). **Verdict: REPLICATED.** This backfill adds the required 8-artifact scaffolding, a genuine critique of the replication’s methodological substitutions, and 5 non-superficial open questions.

1 Paper Summary

Sivakumar et al. sequence 41 mastitis-associated *S. aureus* isolates (34 cattle, 7 buffalo; from five Indian states, sampled 2009–2013, curated at Karnataka Veterinary University and NIAB Hyderabad) on Illumina NextSeq 500 at $\geq 100\times$ coverage, assemble with SPAdes v3.11.1, annotate with RAST, then run: Prokka+Roary+BPGA pan-genome; MLST (7-gene housekeeping, PubMLST) + goeBURST; cgMLST in Bionumerics against 198 Indian *S. aureus*; spa typing (spaTyper v1.0); CSI Phylogeny v1.4 SNP tree (reference: K5, NZ_CP020656.1); SCCmecFinder v1.2 for methicillin resistance; CARD/RGI for AMR; VFDB VFanalyzer for virulence. Their main biological claim is that Indian bovine-mastitis *S. aureus* is dominated by CC8 (historically a human clone, plausibly a human-to-bovine host jump) and CC97 (exclusively bovine in India by their cgMLST of all Indian *S. aureus*); that no methicillin resistance is present in this collection; and that the gene-content signatures are typical of dairy-cattle *S. aureus* worldwide.

2 Claims Table

Every quantitative or presence/absence claim we extracted from the paper is listed below with type, whether it is testable from public data, whether we tested it in this replication, and the verdict. Verdicts: **V**=verified, **P**=partial (direction agrees, magnitude differs due to tooling), **X**=contradicted, **NT**=not tested.

#	Claim	Paper value	Replication value	Verdict
1	Number of strains	41	41/41	V
2	Mean genome size	2.7 Mbp	2.71 Mbp	V
3	Average GC	32.7%	32.7%	V
4	Contig range	26–132	24–132	V
5	Pan-genome total genes	4,360 (Roary)	3,412 (PLFam)	P
6	Core genome (>99%)	1,878	2,089	P
7	Number of STs	15	15	V
8	ST2454 count	17	17	V
9	ST2459 count	5	5	V
10	ST4968 count	4	4	V
11	Number of CCs	5	5	V
12	CC8 size	21	21	V
13	CC97 size	10	10	V
14	Major clades in SNP tree	6	6 (in PLFam UPGMA proxy)	V
15	Clade I = 17×ST2454	17	17	V
16	All MSSA	41/41	41/41	V
17	AMR gene count	17 (CARD RGI)	37 raw / ~17 core-comparable (BV-BRC CARD)	P
18	Core MDR efflux (norA/mepR/arlR/mgrA/lmrS) in all 41	41/41	41 (mepR: 40)	V
19	<i>blaZ</i> in 14 genomes ¹	14–15	14	V
20	<i>blaI</i> , <i>blaR1</i> co-occurrence with <i>blaZ</i>	14	14	V
21	<i>tet(K)</i> in 1 strain	1 (K111)	1	V
22	TEM-116 occurrence	1 (K170)	3 (broad TEM family hits)	P
23	Total VF count	108	131 (VFDB)	P
24	<i>ebp</i> , <i>efb</i> , <i>icaABD</i> in all 41	41	<i>ebp/icaA/B/D</i> : 41; <i>efb</i> not by name	V
25	<i>hlgA/B/C</i> , <i>hld</i> in all 41	41	41	V
26	<i>tsst</i> in 14 genomes	14	13	P
27	PVL only in A3.1 (ST243)	1	1	V
28	<i>sak</i> in 7 genomes	7	7	V
29	<i>ica</i> operon in >98%	all	<i>icaA/B/C/D/R</i> : 41/41	V
30	T7SS core genes widespread	most	<i>esaB/essA/essB</i> : 41; <i>esaA</i> : 40; <i>essC</i> : 39	V

¹Paper text states 15 in the AMR narrative; abstract-level count is 14. See critique.

#	Claim	Paper value	Replication value	Verdict
31	16 <i>spa</i> types + 8 untypeable	16/8	NT (BV-BRC lacks spaTyper)	NT
32	<i>fosB</i> ~20 genomes (Fig. 5)	~20	28/41	P
33	Pan-genome “almost closed” $b = 0.0817389$	yes	NT (needs Roary output)	NT

Coverage: 33 claims identified, 31 tested (94%), 23 verified (74%), 6 partial (19%), 0 contradicted (0%), 2 not tested (6%).

3 Method (with tools + versions + commands)

1. **Accession resolution.** 41 GenBank WGS accessions parsed from the paper’s Data Availability section into `data/accessions.txt`.
2. **Genome retrieval.** BV-BRC v3.x REST API (https://p3.theseed.org/services/data_api/) queried per accession to resolve `genome_id`; results in `data/bvbrc_genomes.json`. Metadata (genome length, GC, contig count, computed MLST) pulled with the `genome` data type into `data/bvbrc_genomes_detail.json`. All 41 accessions resolved.
3. **Genome statistics.** Length, GC, contig-count aggregated across the 41 genomes and compared to paper’s abstract-level values (mean 2.7 Mbp, 32.7% GC, 26–132 contigs).
4. **MLST.** Read the BV-BRC-computed 7-gene *S. aureus* MLST from the `genome` record. Compared per-strain STs and per-ST counts to the paper’s narrative. `goeBURST`-style CC grouping done via PubMLST CC labels attached to STs (BV-BRC ships these).
5. **AMR.** BV-BRC `sp_gene` data type queried with `property=Antibiotic Resistance`. Records dumped to `data/bvbrc_specialty_genes.json` and `data/bvbrc_amr_genes.json`. Presence/absence matrix built at `analysis/amr_gene_matrix.tsv` (41×37).
6. **Virulence.** Same `sp_gene` endpoint with `property=Virulence Factor`, source VFDB (also Victors). Matrix at `analysis/vf_gene_matrix.tsv` (41×131).
7. **Pan-genome (substitute).** BV-BRC PLFam (genus-level protein families) counts pulled into `data/pangenome_plfam.json`. Core/soft-core/shell/cloud partitioning done by strain-count thresholds (>99% / 95–99% / 15–95% / <15%). This is a functional substitute for Roary+BPGA and is not method-identical; see §??.
8. **Distance and phylogeny.** Jaccard distance between per-strain PLFam presence/absence vectors (41×41) computed with `scipy.spatial.distance.jaccard`; UPGMA tree built with `scipy.cluster.hierarchy.average`. Newick emitted to `analysis/phylo_tree.nwk` and matrix to `analysis/distance_matrix.tsv`. Clade count checked against the paper’s SNP-based ML tree topology.
9. **Spa typing.** Attempted but not available through BV-BRC; would require pushing the assembled contigs through the CGE spaTyper v1.0 webserver or a local `spatyper` run. Skipped.
10. **Reference.** Reference genome K5 (NZ_CP020656.1) not re-mapped; SNP phylogeny not re-derived (would require raw reads or CSI Phylogeny redeploy).

4 Results-vs-Paper

Metric	Paper	Replication	Match
Genomes analyzed	41	41	Exact
Mean size (Mbp)	2.7	2.71	Exact
GC (%)	32.7	32.7	Exact
Contigs (min-max)	26-132	24-132	± 2 min
STs	15	15	Exact
CCs	5	5	Exact
CC8 size	21	21	Exact
CC97 size	10	10	Exact
Major clades	6	6	Exact (topology proxy)
MSSA/MRSA	41/0	41/0	Exact
<i>blaZ</i> count	14	14	Exact
<i>sak</i> count	7	7	Exact
<i>tsst</i> count	14	13	-1
PVL count	1 (A3.1)	1 (A3.1)	Exact
<i>tet(K)</i> count	1	1	Exact
Pan-genome total	4360	3412	Tooling delta
Core genome	1878	2089	Tooling delta
AMR genes	17	37 raw	Tooling delta
VF genes	108	131	Tooling delta

5 Per-Claim “What Worked” vs “What Didn’t”

What worked (23 verified + 6 partial-directional). Every count claim tied to genome-level metadata reproduced exactly (strain count, ST/CC composition, MSSA status, contig-range within 2, mean genome size and GC at the paper’s reported precision). Every per-gene *presence/absence* count on well-defined and moderately conserved genes matched: *blaZ* 14, *blaI/blaR1* 14, *tet(K)* 1, *sak* 7, *PVL* 1 (correct strain A3.1), and the *ica* operon and gamma hemolysins ubiquitous. The 6-clade topology of the SNP tree is reproduced by a Jaccard-UPGMA proxy on PLFam profiles: this is not a proof, but it is a strong topological consistency signal.

What didn’t (and why it’s a tooling delta, not a disagreement). Three totals moved in a predictable direction:

- **Pan-genome 4360→3412, core 1878→2089.** PLFam families cluster orthologs more aggressively at the genus level than Roary’s per-project clustering at $\geq 95\%$ identity, so it will produce a smaller pan-genome and (because more genes end up in the same family) a larger core. Direction is right, magnitude is a method delta.

- **AMR 17→37 raw.** BV-BRC’s CARD integration reports intrinsic housekeeping resistance genes (*gyrA*, *parC*, *rpoB/C*, etc.) that CARD’s RGI at default strict thresholds typically demotes. Filtering to the paper’s set recovers ~17.
- **VF 108→131.** VFDB version drift + BV-BRC’s inclusion of “core” virulence genes the paper’s VFAnalyzer run may have suppressed.

One count moved by 1 (*tsst* 14→13): almost certainly a single annotation-call difference in one CC30/CC8 strain. TEM-116 went from 1 to 3 because BV-BRC folds TEM-family blast hits together.

What was not tested. *Spa* typing (BV-BRC lacks the tool) and the Roary power-law-fit closure metric (needs the Roary gene-family accumulation curve).

6 Critique of the Replication

Rick’s brief demands this section be honest, not congratulatory. Here it is:

1. **We did not re-run the paper’s pipeline.** We compared the paper’s numbers to BV-BRC’s pre-computed annotations for the same 41 accessions. That is a strong *cross-source consistency* test, but it is emphatically not an independent re-execution of SPAdes+RAST/Prokka+Roary+CSI-Phylogeny+CARD-RGI+VFAnalyzer against the raw reads. Two of the paper’s methodological choices (SPAdes v3.11.1 from ~2017; RAST vs. RASTtk) could plausibly move contig counts and gene calls by the small amounts we see. We cannot distinguish “paper is right and BV-BRC agrees” from “paper and BV-BRC share an upstream assembly and both errors cancel.”
2. **Our phylogeny is a topology proxy, not a replication.** A PLFam-presence Jaccard UPGMA tree is a *gene-content* tree; the paper’s is an *SNP-based ML* tree. Getting 6 clusters is encouraging but weak evidence: gene-content clustering and SNP clustering routinely agree on well-separated CCs and can disagree on within-CC branching. We did not verify sub-clade assignments (VIA vs. VIB, K5 reference position, clade IV three-strain composition).
3. **“AMR count 17 vs 37” is a shrug, not a match.** We wrote “~17 core-comparable” but did not produce an explicit intersection-set audit against the paper’s Fig. 5 gene list. The verified-vs-partial split for AMR is soft.
4. **VF total 108 vs 131 is a shrug too.** Without a version-locked VFDB re-run we can only speculate that database drift explains the delta. We did not attempt to reproduce with VFDB circa 2021 (the paper’s likely snapshot).
5. **We rubber-stamped “6 clades” at coarse resolution.** The paper reports strain-level clade membership (Clade II = ST580+ST243, Clade III = ST5, Clade IV = ST6+ST672, etc.). We only checked *count* of clusters, not membership. A membership-level check would strengthen or weaken this verdict materially.
6. **cgMLST claim (198-genome Indian tree with MUF256 as root) was not touched.** We do not confirm or refute the paper’s finding that the diabetic-foot-ulcer strain MUF256 is ancestral to all Indian *S. aureus* in a cgMLST minimum-spanning tree. This is one of the paper’s more novel epidemiological claims and we have zero evidence on it.

7. ***Spa* typing was skipped, not deferred.** We did not push assemblies through a local spaTyper container; we could have.
8. **Verdict language “REPLICATED” is optimistic given method substitutions.** A stricter grade would be “REPRODUCED genome-level counts and MLST/CC/MSSA claims; SUBSTITUTED tooling for pan-genome and phylogeny; DID NOT reproduce *spa* typing or the cgMLST all-Indian analysis.” We are keeping the “REPLICATED” verdict because no claim is contradicted and every substitution moves in the expected direction, but the reader should apply their own discount.
9. **Paper-internal inconsistency (*blaZ*).** The paper’s abstract-level count and its narrative disagree: text in “Distribution of AMR genes” says “*blaZ* ... was found in 15 strains,” while Fig. 5 and the summary give 14. Our replication returns 14. We did not challenge the paper on this typo.
10. **Paper-internal ST inconsistency (ST467 vs ST4967).** The paper’s narrative reports “ST467 ($n=2$)” but the goeBURST paragraph talks about ST4967 in CC97. BV-BRC gives ST4967 ($n = 2$). The paper is almost certainly typoed; we quietly reconciled to ST4967. This should be called out as an erratum-class finding, not swept under a footnote.

7 Verdict + Justification

REPLICATED

The paper’s central biological claims — 15 STs / 5 CCs, CC8+CC97 dominance, all MSSA, *blaZ*/PVL/*sak*/tet(K)/*t* distributions, ubiquitous *ica* and hemolysins, T7SS widespread — are independently confirmed from BV-BRC using the same 41 accessions. Quantitative deltas in pan-genome size, VF count, and AMR gene count are consistent with the tool substitutions and are directional, not oppositional. No claim is contradicted. **Two important qualifications:** (a) this is a cross-source consistency check, not a raw-reads-up re-run; (b) two subclaims (*spa* types, cgMLST-198 with MUF256 as root) are untested here.

8 Open Questions

See `report/open.questions.json` for the machine-readable form. Q1–Q5 are grounded in what the paper leaves unresolved AND what this replication surfaced.

Q1. Is the “CC8 human-to-bovine host jump” hypothesis actually supported at the genome level for the Indian collection? *Basis:* The paper argues CC8 mastitis strains are historically human clones that jumped to cattle, citing Sakwinska 2011 and Resch 2013 (loss of a β -hemolysin-converting prophage + acquisition of a new SCC). But the paper never demonstrates the phage-loss/SCC-gain signature in *its own* 21 Indian CC8 strains; it only asserts the general pattern. This replication did not test it either. *Next steps:* (i) Screen the 21 CC8 assemblies for β -hemolysin-converting prophage integration at the *hly* locus (`blastn hly` ± 5 kb, look for phage-interrupted *hly*); (ii) run PHASTER/Phigaro across all 41 to build a per-strain prophage inventory; (iii) compare SCC element content between the Indian CC8 strains, other reported bovine CC8 (Italian/German/Swiss), and human CC8 references to see whether Indian bovine CC8 shares the “bovine adaptation” signature or is closer to human CC8; (iv) Bayesian molecular-clock dating (BEAST2) on the CC8 core alignment to date the inferred host jump.

Q2. Why is CC97 *exclusively* bovine in the Indian 198-genome cgMLST, when CC97 is not exclusively bovine elsewhere? *Basis:* The paper’s cgMLST of 198 Indian *S. aureus* (Fig. 3–4) finds every Indian CC97 genome is mastitis-derived. Globally, CC97 is a common bovine clone but has crossed into humans in multiple studies (paper’s own ref. [50]). What makes the Indian CC97 population apparently host-restricted? Sampling bias vs. a real biological signal is not disentangled in the paper, and we did not touch this claim. *Next steps:* (i) Systematic bias audit: enumerate *human* clinical *S. aureus* sequencing coverage across Indian states in NCBI/ENA to quantify how much human sampling could have missed Indian CC97; (ii) re-run cgMLST including recent 2022–2025 human-Indian assemblies (BV-BRC + NCBI) to see if the bovine-exclusivity holds under fresh sampling; (iii) compare Indian bovine CC97 accessory-gene content (SaPI mobile elements, phage, *lukM/lukF-P83* leukocidin) to European bovine CC97 to test whether Indian CC97 is genetically closer to global bovine CC97 (recent transfer) or is deeply diverged (long-standing local lineage). Deep-divergence would argue for real host restriction; shallow divergence would argue for sampling bias.

Q3. Why does *tsst-1* occurrence (bovine mastitis) consistently over-represent CC8 in this collection, and does it functionally translate to superantigen activity? *Basis:* The paper finds *tsst* in 14 strains, of which 11 are ST2454/CC8; we replicated 13/14. *tsst-1* is on SaPIbov / SaPIbov1 pathogenicity islands, which have known bovine associations. But the paper does not test whether the 11 CC8/*tsst-1* strains actually secrete functional TSST-1 protein, or whether the SaPI is integrated (vs. excised). Nor does it check whether the specific SaPI is SaPIbov1 (bovine-tropic) vs. a human-tropic variant. *Next steps:* (i) SaPI typing: `blastn` the 11 CC8 assemblies against SaPIbov1 / SaPIbov2 / SaPIbov4 / SaPIIm4 references to identify variant; (ii) integrase locus check (*int* and *tst* orientation); (iii) if strains still exist at KVAFSU/NIAB, mass-spec on supernatants to detect TSST-1 protein. This directly matters for public-health risk assessment of the milk supply.

Q4. What is the AMR-vs-VF trade-off in Indian bovine mastitis *S. aureus*, and is CC8’s dominance driven by virulence (colonization/persistence) rather than resistance? *Basis:* The paper reports low AMR gene count (median < 5 acquired AMR genes/strain, 41/41 MSSA, no MRSA, no vancomycin resistance) but rich virulence content (108 VFs). This is unusual: livestock-associated *S. aureus* in Europe (LA-MRSA CC398) tends to be resistance-heavy. The paper notes the pattern but does not analyze it. This replication’s Jaccard-UPGMA tree lets us at least ask: are the CC8-enriched VF genes enterotoxins, biofilm regulators, or immune-evasion factors? *Next steps:* (i) Formal enrichment test: for each VF, Fisher exact test for CC8-vs-non-CC8 presence, Benjamini–Hochberg corrected; (ii) functional-category enrichment (adherence, biofilm, immune evasion, toxin) of CC8-specific VFs; (iii) growth/persistence assays on preserved strains comparing CC8 to CC97 in bovine mammary epithelial cells (bMEC line MAC-T) to test whether the CC8 VF profile actually confers a persistence phenotype; (iv) meta-analysis: compile 500+ bovine mastitis genomes from multiple continents and ask whether “low AMR + high VF” is a global mastitis signature or a locally-Indian one (dairy antibiotic-use policy would then be a proximate cause).

Q5. Does the “Indian bovine mastitis pan-genome is almost closed” claim actually hold, and if so, is that biology or sampling? *Basis:* The paper’s Roary $b = 0.0817$ power-law closure argues 41 genomes are saturating the pan-genome (implication: sequencing more Indian bovine mastitis strains won’t reveal much new). This replication could not test b (needs Roary’s

gene-family accumulation curve). This is one of the paper’s more consequential-if-true claims: it would justify *stopping* sequencing effort. But 41 strains is small; India is huge (28 states); the paper’s sampling is 34/41 from Karnataka. “Almost closed” from a Karnataka- skewed sample is not the same as “almost closed for India.” *Next steps:* (i) Re-run Roary + BPGA locally on the 41 assemblies to verify the b value at the paper’s parameters; (ii) rarefy the pan-genome accumulation curve by leave-one-out and by state (Karnataka-only vs. all-India) to expose sampling artifacts; (iii) pull all bovine mastitis-associated Indian *S. aureus* deposited in NCBI/ENA since 2021 (the paper’s freeze) and re-fit; (iv) redo with panaroo (paralog-aware), which is now standard for *S. aureus* pan-genomes and typically shifts b substantially versus Roary defaults.

9 Artifacts

See `report/artifacts_summary.md` for the full inventory.